cylcle, the targeted sequences on both strands have been copied. When the three – step cycle is repeated, the four strands from the first cycle are copied to produce eight fragments. The third cycle yields 16 products. Theoretically, 20 cycles will produce about one million copies of the target DNA sequence. Each cycle of PCR takes about 3 – 5 minutes.

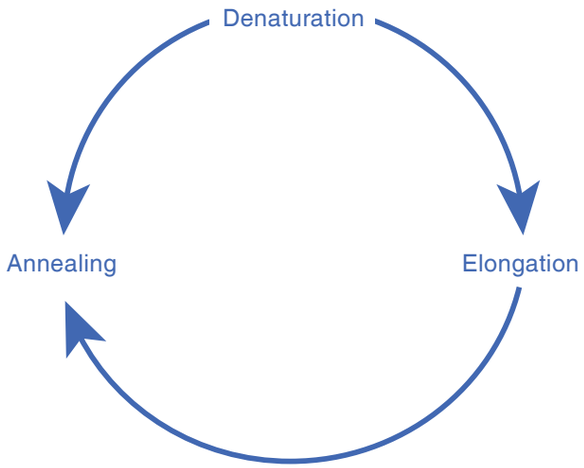


Figure 12.32: Three steps PCR cycle

The PCR technique has now been automated and is carried out by a specially designed machine (Figure 12.33) PCR machines are now fully automated and microprocessor controlled. They can process up to 96 samples at a time. PCR machines can carry out 25 cycles and amplify DNA 10^5 times in as little as 57 minutes.

The PCR has many applications in research and in commercial arena, including generating specific DNA segments for cloning or sequencing, amplifying DNA to detect specific genetic defects, and amplifying DNA for fingerprinting in crime scene investigation.

PCR technology is improving continually. Various forms of PCR are available. RNA too can be efficiently used in PCR procedures. Cellular RNAs and RNA viruses may be studied even when the RNA is present in very small amounts (as few as 100 copies can be transcribed and amplified). Quantitative PCR is quite valuable in virology and gene impression studies. PCR is modified as per the specific demands of the situation. Thus there are many variations in the original PCR Examples nested PCR, inverse PCR, reverse transcription PCR, time quantitative PCR, RAPD, RFLP, AFLP.

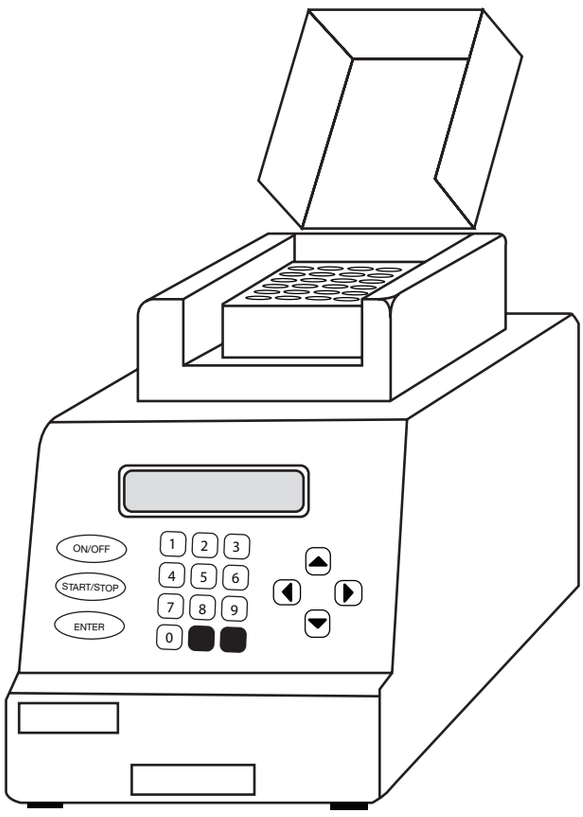


Figure 12.33: PCR machine

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